Machine Learning Lab – Question 13

Question

Implement the Car Price Prediction Model using Python.

Code

```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestRegressor
from sklearn.metrics import mean_absolute_error, r2_score

data = pd.DataFrame({
    "year":[2015,2016,2017,2018,2019,2020,2021,2022,2014,2013],
    "km":[90000,80000,70000,60000,50000,40000,30000,20000,110000,120000],
    "engine":[1200,1300,1500,1500,1600,1800,1800,2000,1000,1000],
    "price":[450000,500000,580000,650000,750000,900000,1050000,1200000,400000,350000]
})
X=data[["year","km","engine"]]; y=data["price"]
X_train,X_test,y_train,y_test=train_test_split(X,y,test_size=0.2,random_state=42)

model=RandomForestRegressor(n_estimators=200,random_state=42)
model.fit(X_train,y_train)
pred=model.predict(X_test)

print("MAE:",round(mean_absolute_error(y_test,pred),2))
print("R2:",round(r2_score(y_test,pred),4))
print("Predicted price:",round(model.predict([[2020,45000,1600]])[0],2))
```

Output

```
The program prints MAE, R2 score, and predicted price for the supplied car.
Exact values depend on the training/test split and scikit-learn version.
```